

Remineralization and You: How to Maintain Strong Teeth

You know the basics: that for healthy teeth – i.e., teeth without cavities – you should brush twice a day, floss often, and not have too much sugar.

But taking a step back to consider the various factors producing strong teeth can give you a much better understanding of what's going on in your mouth, and how to maintain a truly healthy smile.

Teeth are composed of the minerals Calcium and Phosphate. In short, teeth are made up of several layers of more-or-less calcified tissue, covering an inner 'pulp' of nerve endings. The outermost protective layer is the tooth enamel – *the hardest substance in the body* – which shelters the softer 'dentin' tissue within. (Think of it a bit like a buff dude protecting his nerdy friend.) It's when the enamel is worn away, and the sensitive inner tissue exposed, that cavities may occur. (The buff dude is injured, so the nerd gets hurt.)

Fundamentally, then, there are two ways to protect against cavities: to prevent the enamel's 'dimineralization', and to foster the tooth's 'remineralization'. Yes, our calcium-and-phosphate laden enamel can be worn down. But the tooth can also be fortified in various ways, including by the production of 'secondary dentin' by specialized 'odontoblast' cells in the pulp. (We can prevent the buff dude from becoming injured, or help him keep working out, and in some cases help the nerd heal... – okay, this analogy ends here. You're welcome.)

Strategies for preventing *dimineralization* will be familiar. We brush our teeth and floss to clear away bacteria (or food particles that feed bacteria) that, left unchecked, will produce acid that damages our enamel. We limit foods and drinks that are themselves particularly acidic (soda, juice), or that are sugary enough to set off a bacterial feeding frenzy. And when we brush, we do it gently enough to avoid physical erosion.

Remineralization is a bit more interesting. The process is unfortunately not as simple as popping Calcium and Phosphorous supplements like they're Tic Tacs.

Three factors contribute to our body's ability to remineralize our teeth.

1. The presence of enough minerals in the diet.
2. The presence of enough vitamins in the diet that allow our body to process the above minerals.

3. Your body's capacity for absorbing and releasing the above minerals and vitamins.

Let's look at all three.

1. **The presence of enough minerals in the diet.**

Simply put, satisfying this factor entails consuming more Calcium-and-Phosphorous-rich foods. The mineral Zinc has also been shown to help with tooth remineralization.

Good foods for this purpose are: Broccoli, Kale, Cabbage, Tofu, Almonds, Pumpkin Seeds, Shellfish, and Red Meat.

Mineral supplements also fall within this category.

2. **The presence of enough vitamins in the diet that allow our body to process the above minerals.**

The two most important vitamins for processing the above minerals are Vitamin D and Vitamin K2. Vitamin D increases the body's absorption of calcium and phosphorous, and Vitamin K2 helps direct calcium toward the teeth and bones, rather than the kidneys and arteries.

The best sources of Vitamin D are exposure to sun, fatty fish like tuna, fermented cod liver oil, egg yolks. The best sources of Vitamin K2 are egg yolk, fermented veggies, and chicken livers. Yeah, you might want to take a supplement for that one.

Vitamins A and E can also help in the absorption of minerals.

3. **Your body's capacity for absorbing and releasing the above minerals and vitamins.**

Interestingly, your body's ability to process the above minerals and vitamins can depend on your consumption of other foods – namely, those containing 'phytic acid'.

Phytic Acid is a form of phosphorous not easily absorbed by humans. It's actually how plants supply their own phosphorous – a supply that's adapted to be difficult to digest by animals to help seeds pass through their stomachs intact and ready to propagate. What's worse is that these 'phytates' leach calcium and other minerals along the way.

Phytates may have other benefits, however – not to mention that most people aren't willing to give up grains just yet. Instead, you can at least reduce your consumption, which practically speaking mostly just means laying off the gluten.

So, big picture: strong healthy teeth can be achieved not only by preventing *dimineralization*, with sensible oral hygiene practices and reduced sugar intake, but by *remineralizing*. This means a diet high in calcium and phosphorous, high in Vitamin D and K2, and low in gluten. Happy Bruushing.